

NIRIKSHAK

Frontend, UI Flow & Backend/ML Data Contract

Prototype Specification for SIH 2026 • Problem Statement SIH26184

Prepared specifically for the Frontend / GIS / LEA Interface team

Primary demo journey	Complaint -> Prediction -> Location -> Alert -> LEA -> Bank -> Action -> Outcome
Primary frontend objective	Make the predictive result operationally understandable and actionable.
Map objective	Telangana -> District -> Area -> ATM/location drill-down with observed vs predicted information clearly separated.
Prototype data	Use dummy/synthetic records with the same contract as the future integrated backend.
Important constraint	This document does not prescribe ML/model implementation; it defines what the UI receives and displays.

Source basis

This specification is derived from the uploaded six-page SIH idea deck, especially its proposed workflow, technical architecture, GIS/alert layer, stakeholder roles, feasibility constraints, and impact flow. The deck describes Nirikshak as a system that converts cybercrime complaints into time-aware, probabilistic cash-out forecasts and delivers ranked intelligence to LEAs, banks/FIs and I4C.

Prototype principle

The frontend should resemble an operational government intelligence platform, not a generic AI analytics dashboard. The map is important, but the core demonstration is the conversion of a complaint into a predicted cash-out location and then into an accountable response workflow.

1. What the Frontend Must Prove

The judges should be able to understand the entire value chain without seeing the underlying model implementation.

- >	2 Financial context	- >	3 Prediction	- >	4 Risk location	- >	5 Alert	- >	6 LEA / Bank action	- >
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The prototype must answer six questions at every stage:

- **What happened?** — complaint/case and known transaction context.
- **Where may the money cash out?** — district, area and specific ATM/location.
- **When?** — predicted withdrawal window, with timezone and freshness.
- **Why this location?** — supporting transaction path, convergence and evidence indicators.
- **Who needs to act?** — responsible LEA, bank/FI and I4C coordination view.
- **What happened after the alert?** — acknowledgement, assignment, action and outcome.

The uploaded deck already establishes the Predict -> Prioritize -> Coordinate -> Intervene story. The prototype should extend this to include explicit action recording and outcome feedback so that the operational loop is visible.

2. Product Structure

Product / View	Primary user	Purpose
I4C Command Center	I4C officer	State/national coordination, risk overview, active alerts and cross-jurisdiction visibility.
LEA Investigation Console	LEA officer/investigator	Case investigation, prediction detail, GIS drill-down, alerts and action recording.
Bank/FI Response Console	Bank/FI officer	Receive relevant intelligence, review affected account/transaction context and record response.
Citizen / Complaint View	Citizen / victim	Submit/track complaint and see non-sensitive case status only.
Shared Alert & Intelligence components	All authorized roles	Alert lifecycle, prediction cards, map layers, transaction path, notifications and audit timeline.

3. Roles, Inputs and Visibility

The same platform should use role-based access. Do not create four visually unrelated applications. The role determines which data and actions are available.

Role	Inputs from role	Frontend receives / can view	Primary actions
Citizen / Victim	Complaint details; contact/identifier; incident time; financial loss details; optional transaction references.	Complaint ID, acknowledgement, status, limited non-sensitive updates.	Submit complaint; track status; provide requested follow-up information.
LEA	Investigation notes; action updates; acknowledgement; assignment; outcome/verification.	Case intelligence, transaction path, predicted locations, GIS, alerts, evidence metadata, audit trail.	Acknowledge, assign, investigate, share/notify, record action, record outcome.
Bank / FI	Alert acknowledgement; verification result; response/action; outcome.	Only relevant alerts/cases/accounts/transactions according to authorization; predicted location when operationally relevant.	Acknowledge, verify, hold/escalate according to workflow, add response, close/resolve.
I4C	Coordination decisions; escalation; cross-state/cross-LEA notes; outcome review.	Aggregated intelligence, all authorized alerts, map, case drill-down, coordination status, audit.	Coordinate, escalate, assign, review, monitor outcomes, inspect system-wide patterns.

Sensitive information rule

The frontend must not assume that every role gets the same transaction graph, account identifiers or evidence. The backend should enforce authorization; the UI should also hide unavailable actions and fields so the prototype communicates realistic RBAC.

4. Golden Demo User Flow

available	- >	4 District risk	- >	5 ATM risk	- >	6 Alert generated	- >	7 LEA opens	- >	8 Bank notified
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5. Complete Screen Inventory

Build the following screens. Screens marked ★ are the minimum set for the screening-round walkthrough.

ID	Screen	Role	Priority	What must be visible
S01	Secure Login / Role Selection	All	High	Brand, role, authentication state, secure-session indicator.
S02	Citizen Complaint Submission	Citizen	Medium	Incident details, financial loss, transaction reference, district/location if known, consent/acknowledgement.
S03 ★	Complaint / Case Created	Citizen + LEA	Very High	Case ID, status, timestamp, next workflow state.
S04 ★	LEA Dashboard	LEA	Very High	Active cases, high-risk predictions, active alerts, pending actions, mini-map.
S05 ★	I4C Command Center	I4C	Very High	State overview, risk map, alert queue, coordination status.
S06 ★	Telangana GIS Overview	LEA/I4C	Very High	District polygons, district risk, ATM dot layer, filters, time state.
S07 ★	District Drill-down	LEA/I4C	Very High	Area polygons/shades, ATM points, district aggregation and threshold highlight.
S08 ★	ATM / Location Detail	LEA/I4C/Bank	Very High	Risk, ATM details, coordinates, Google Maps link, withdrawal history, related cases.
S09 ★	Prediction / Intelligence Detail	LEA/I4C	Very High	Predicted window, risk, confidence, supporting paths, freshness, explanation indicators.
S10 ★	Transaction / Fund-flow Path	LEA/I4C	Very High	Victim -> accounts -> predicted ATM path, amounts/timestamps where authorized.
S11 ★	Alert Detail	LEA/Bank/I4C	Very High	Alert lifecycle, target location, case, prediction, recipients, actions.
S12	LEA Investigation Workspace	LEA	High	Case timeline, intelligence, notes, assignments, evidence metadata, actions.
S13	Bank/FI Response Queue	Bank/FI	High	Incoming alerts, severity, SLA/status, acknowledgement and response controls.
S14	Bank/FI Alert Detail	Bank/FI	High	Relevant transaction/account context, prediction, response workflow.
S15	I4C Coordination View	I4C	High	LEA/bank recipients, acknowledgement, escalations, status.
S16	Audit / Activity Timeline	LEA/I4C/Bank	High	Who did what, when, on which alert/case.
S17	Citizen Complaint Status	Citizen	Medium	Safe status milestones only.
S18	Notifications Center	All authorized	Medium	Unread alerts, acknowledgements, escalations and system notices.

6. Backend -> Frontend Data Contract

The frontend team should build against stable JSON contracts. Dummy data should use exactly the same field names and nesting that the integrated backend will eventually return.

Recommended common envelope

```
{
  "success": true,
  "request_id": "REQ-20260916-00091",
  "generated_at": "2026-09-16T13:32:18Z",
  "data_version": "v1",
  "data": {}
}
```

Common metadata fields

Field	Type	Meaning
success	boolean	Whether request succeeded.
request_id	string	Traceable request identifier.
generated_at	ISO datetime	When the response was produced.
data_version	string	Frontend contract version.
as_of	ISO datetime	Point-in-time boundary for intelligence, when relevant.
freshness_seconds	number	Age of the intelligence relative to the current system time.
scope	object	State/district/area/organization scope applied to the result.

Important distinction: `generated_at` is not the same as `as_of`. The latter tells the UI what information was considered available at the intelligence snapshot. This supports the deck's point-in-time and refresh/traceability concept.

7. Core Data Objects

Object	Required for frontend	Used by
Case / Complaint	Yes	Citizen, LEA, I4C
District Risk	Yes	GIS, I4C, LEA
Area Risk Polygon	Yes	District drill-down map
ATM / Cash-out Location	Yes	GIS, prediction, alert, bank/LEA
Prediction	Yes	LEA, I4C, alerts
Predicted Path	Yes	Prediction detail / fund-flow view
Withdrawal History	Yes for ATM detail	ATM detail
Alert	Yes	LEA, Bank, I4C
Action / Outcome	Yes	LEA, Bank, I4C
Audit Event	Yes	LEA, Bank, I4C
Notification	Yes	All authorized roles

8. GIS Data Contract & Interaction Rules

The GIS must be hierarchical: Telangana -> District -> Area -> ATM/location. All ATMs may be visible as small dots at state level, but they become operationally clickable only after the relevant district is drilled into.

8.1 Telangana overview response

```
{
  "map_scope": "TELANGANA",
  "districts": [
    {
      "district_id": "D-HYD",
      "district_name": "Hyderabad",
      "risk_level": "HIGH",
      "risk_score": 0.84,
      "atm_total": 86,
      "high_risk_atm_count": 9,
      "highlight_threshold": 5,
      "district_highlight": true,
      "geometry": {
        "type": "Polygon",
        "coordinates": [
          "<district-boundary-coordinates>"
        ]
      }
    }
  ],
  "atm_overview": [
    {
      "atm_id": "ATM-HYD-00421",
      "district_id": "D-HYD",
      "lat": 17.385,
      "lon": 78.4867,
      "map_interaction": "DOT_ONLY"
    }
  ]
}
```

8.2 District highlight threshold

For the prototype, use a configurable threshold rather than hardcoding the number 5 into the UI. Example: **highlight_threshold = 5**. The backend returns the computed count and boolean. The UI only renders the rule. This avoids turning a prototype display threshold into an unexplained business rule.

Field	Type	UI use
district_id	string	Stable identifier.
district_name	string	Map label and filter.
risk_score	number 0–1	Color/intensity and numeric display.
risk_level	LOW/MEDIUM/HIGH/CRITICAL	Legend and badge.
atm_total	integer	District summary.
high_risk_atm_count	integer	Threshold calculation.
highlight_threshold	integer	Display/config context.
district_highlight	boolean	Whether district gets highlighted.
geometry	GeoJSON	Actual district polygon.
predicted_window_start/end	ISO datetime	Temporal filter/summary if district-level forecast exists.

8.3 Area-level risk polygons

```
{
  "district_id": "D-HYD",
  "areas": [
    {
      "area_id": "A-HYD-017",
      "area_name": "Area Alpha",
      "risk_score": 0.91,
      "risk_level": "CRITICAL",
      "high_risk_atm_count": 4,
      "geometry": {
```

```

        "type": "Polygon",
        "coordinates": [
            "<actual-area-polygon>"
        ]
    },
    {
        "area_id": "A-HYD-022",
        "area_name": "Area Beta",
        "risk_score": 0.62,
        "risk_level": "MEDIUM",
        "high_risk_atm_count": 1,
        "geometry": {
            "type": "Polygon",
            "coordinates": [
                "<actual-area-polygon>"
            ]
        }
    }
]
}

```

Do not create radial circles around ATMs as the primary risk visualization. The map should use actual geographic/administrative polygons wherever the underlying boundary data exists. ATM markers then sit on top of those areas.

9. ATM / Cash-out Location Data Contract

At state level, ATM points can be rendered as non-clickable dots. After district drill-down, the frontend can activate the detailed ATM layer and allow selection.

```
{
  "atm_id": "ATM-HYD-00421",
  "name": "Example ATM - Location 421",
  "bank_name": "Example Bank",
  "bank_id": "BANK-014",
  "lat": 17.385044,
  "lon": 78.486671,
  "address": "Example address",
  "district_id": "D-HYD",
  "district_name": "Hyderabad",
  "area_id": "A-HYD-017",
  "area_name": "Area Alpha",
  "risk": {
    "risk_score": 0.91,
    "risk_level": "CRITICAL",
    "rank_in_district": 1,
    "predicted_withdrawal_window": {
      "start": "2026-09-16T14:00:00+05:30",
      "end": "2026-09-16T16:00:00+05:30"
    }
  },
  "map_links": {
    "google_maps": "https://www.google.com/maps/search/?api=1&query=17.385044,78.486671"
  },
  "related": {
    "active_prediction_count": 3,
    "related_case_count": 7,
    "high_risk_path_count": 5
  }
}
```

Field	Type	Required	Displayed
atm_id	string	Yes	Yes
name	string	Yes	Yes
bank_name / bank_id	string	Yes	Yes
lat / lon	number	Yes	Yes
address	string	Preferred	Yes
district / area IDs + names	string	Yes	Yes
risk_score / risk_level	number/string	Yes	Yes
predicted window	datetime range	If predicted	Yes
google_maps	URL	Preferred	Clickable
active_prediction_count	integer	Preferred	Yes
related_case_count	integer	Preferred	Yes
high_risk_path_count	integer	Preferred	Yes

ATM detail must contain three distinct sections:

- **Location** — bank, address, coordinates and external map link.
- **Risk** — score, level, rank, predicted window and freshness.
- **Intelligence** — linked cases, predicted paths and previous withdrawals.

10. Previous Withdrawals Data

```
{
  "atm_id": "ATM-HYD-00421",
  "withdrawal_history": {
    "period": {
      "from": "2026-08-16T00:00:00+05:30",
      "to": "2026-09-16T00:00:00+05:30"
    },
    "summary": {

```

```
    "withdrawal_count": 34,  
    "total_amount": 485000,  
    "avg_amount": 14264.71  
  },  
  "records": [  
    {  
      "withdrawal_id": "W-889201",  
      "timestamp": "2026-09-15T15:02:11+05:30",  
      "amount": 20000,  
      "status": "COMPLETED",  
      "related_case_id": "C-10482"  
    }  
  ]  
}
```

For the prototype, the withdrawal history can be dummy/synthetic. The UI should nevertheless be built around the same structure so the future integration does not require a redesign.

11. Prediction / Intelligence Data Contract

The prediction response is the core object connecting ML output to the operational UI. The frontend should not need to know which model produced it.

```
{
  "prediction_id": "PRED-90021",
  "case_id": "C-10482",
  "status": "ACTIVE",
  "as_of": "2026-09-16T13:24:00+05:30",
  "generated_at": "2026-09-16T13:32:18+05:30",
  "risk": {
    "score": 0.91,
    "level": "CRITICAL",
    "rank": 1
  },
  "predicted_cashout": {
    "location_type": "ATM",
    "atm_id": "ATM-HYD-00421",
    "district_id": "D-HYD",
    "district_name": "Hyderabad",
    "area_id": "A-HYD-017",
    "area_name": "Area Alpha",
    "lat": 17.385044,
    "lon": 78.486671,
    "withdrawal_window": {
      "start": "2026-09-16T14:00:00+05:30",
      "end": "2026-09-16T16:00:00+05:30"
    }
  },
  "supporting_intelligence": {
    "supporting_path_count": 7,
    "converging_path_count": 5,
    "last_observed_transaction": {
      "timestamp": "2026-09-16T13:24:00+05:30",
      "amount": 68000
    }
  },
  "explanation_factors": [
    {
      "label": "Multiple predicted paths converge",
      "value": 5
    },
    {
      "label": "Recent transaction activity",
      "value": "HIGH"
    },
    {
      "label": "Historical cash-out similarity",
      "value": "HIGH"
    }
  ]
},
"freshness_seconds": 492
}
```

Field	Type	Purpose
prediction_id	string	Unique intelligence record.
case_id	string	Connects prediction to complaint/case.
status	string	ACTIVE / EXPIRED / SUPERSEDED / RESOLVED.
as_of	datetime	Point-in-time prediction boundary.
generated_at	datetime	When prediction was published.
risk.score	number 0–1	Numeric risk display; do not imply universal calibration.
risk.level	enum	LOW/MEDIUM/HIGH/CRITICAL.
predicted_cashout	object	ATM/location and time window.
supporting_path_count	integer	Number of supporting predicted paths.
converging_path_count	integer	Paths converging on the predicted ATM.
last_observed_transaction	object	Most recent known event before the snapshot.
explanation_factors	array	Human-readable supporting signals.
freshness_seconds	integer	How old the intelligence is.

UI rule: never label the predicted ATM as a guaranteed withdrawal location. Use language such as Predicted Cash-out Location, Likely Withdrawal Window, and Risk Score. The system supports human decision-making; it does not autonomously determine police action.

12. Predicted Transaction / Fund-flow Path

The deck's central intelligence concept is recursive fund-flow forecasting ending in an ATM/cash-out location. The frontend should expose the path visually, not hide it in a table.

```
{
  "prediction_id": "PRED-90021",
  "path_id": "PATH-07",
  "path_probability": 0.78,
  "nodes": [
    {
      "sequence": 1,
      "node_type": "VICTIM_ACCOUNT",
      "node_id": "MASKED-V-001",
      "label": "Victim Account"
    },
    {
      "sequence": 2,
      "node_type": "ACCOUNT",
      "node_id": "MASKED-A-019",
      "label": "Intermediate Account A"
    },
    {
      "sequence": 3,
      "node_type": "ACCOUNT",
      "node_id": "MASKED-A-044",
      "label": "Intermediate Account B"
    },
    {
      "sequence": 4,
      "node_type": "ATM",
      "node_id": "ATM-HYD-00421",
      "label": "Predicted ATM",
      "lat": 17.385044,
      "lon": 78.486671
    }
  ],
  "edges": [
    {
      "from": "MASKED-V-001",
      "to": "MASKED-A-019",
      "timestamp": "2026-09-16T12:51:00+05:30",
      "amount": 92000
    },
    {
      "from": "MASKED-A-019",
      "to": "MASKED-A-044",
      "timestamp": "2026-09-16T13:10:00+05:30",
      "amount": 71000
    },
    {
      "from": "MASKED-A-044",
      "to": "ATM-HYD-00421",
      "predicted": true
    }
  ]
}
```

Recommended visual hierarchy

1 Victim	- >	2 Account A	- >	3 Account B	- >	4 Predicted ATM
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- Observed edges should be visually distinguishable from the predicted edge.
- Show timestamps and amounts only where the logged-in role is authorized.
- Clicking a path/node should open a compact detail panel, not navigate away unnecessarily.
- Show why the path is included: path probability/score, supporting signals, or convergence count.

13. Case / Complaint Data Contract

```
{
  "case_id": "C-10482",
  "complaint_id": "NCRP-DEMO-10482",
  "created_at": "2026-09-16T12:40:00+05:30",
  "status": "UNDER_INVESTIGATION",
}
```

```

    "crime_category": "CYBER_FINANCIAL_FRAUD",
    "reported_amount": 92000,
    "reporting_timestamp": "2026-09-16T12:40:00+05:30",
    "victim": {
      "display_name": "Masked Victim",
      "contact_masked": "+91-*****421"
    },
    "known_financial_context": {
      "victim_account_masked": "XXXXXX1842",
      "bank_name": "Example Bank"
    },
    "assigned_lea": {
      "unit_id": "LEA-TG-HYD-01",
      "unit_name": "Example LEA Unit"
    }
  }
}

```

14. Alert Data Contract & Lifecycle

An alert is the operational bridge between prediction and intervention. It should be a first-class object, not just a notification toast.

14.1 Alert payload

```

{
  "alert_id": "AL-20481",
  "alert_type": "PREDICTED_CASHOUT",
  "severity": "CRITICAL",
  "status": "DELIVERED",
  "created_at": "2026-09-16T13:32:18+05:30",
  "expires_at": "2026-09-16T16:00:00+05:30",
  "case_id": "C-10482",
  "prediction_id": "PRED-90021",
  "target": {
    "atm_id": "ATM-HYD-00421",
    "district": "Hyderabad",
    "area": "Area Alpha"
  },
  "predicted_window": {
    "start": "2026-09-16T14:00:00+05:30",
    "end": "2026-09-16T16:00:00+05:30"
  },
  "recipients": [
    {
      "type": "LEA",
      "id": "LEA-TG-HYD-01",
      "status": "DELIVERED"
    },
    {
      "type": "BANK",
      "id": "BANK-014",
      "status": "DELIVERED"
    },
    {
      "type": "I4C",
      "id": "I4C-TG-01",
      "status": "VISIBLE"
    }
  ]
}

```

14.2 Lifecycle

- >	2 GENERATED	- >	3 DELIVERED	- >	4 ACKNOWLEDGED	- >	5 ASSIGNED	- >	6 ACTION	- >
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Recommended status set

Status	Meaning
GENERATED	Prediction crossed the alerting rule and an alert record was created.
DELIVERED	Alert reached the intended authorized channel/view.
ACKNOWLEDGED	Recipient explicitly acknowledged receipt.
ASSIGNED	Responsibility assigned to an officer/unit.

Status	Meaning
ACTION_INITIATED	An operational response was recorded.
RESOLVED	Case/alert reached a recorded resolution state.
EXPIRED	Prediction window ended without active alerting relevance.
FALSE_POSITIVE	Outcome feedback indicates the prediction did not materialize.

15. Action / Outcome Data Contract

The UI must make action recording explicit. This is the missing operational step that turns a predictive dashboard into an intervention workflow.

```
{
  "action_id": "ACT-7712",
  "alert_id": "AL-20481",
  "case_id": "C-10482",
  "actor": {
    "role": "LEA",
    "user_id": "OFFICER-102"
  },
  "action_type": "FIELD_VERIFICATION_INITIATED",
  "status": "IN_PROGRESS",
  "created_at": "2026-09-16T13:44:02+05:30",
  "notes": "Local unit assigned for verification.",
  "next_review_at": "2026-09-16T15:00:00+05:30"
}
```

Outcome example

```
{
  "outcome_id": "OUT-1192",
  "alert_id": "AL-20481",
  "outcome_type": "WITHDRAWAL_OBSERVED",
  "recorded_at": "2026-09-16T15:11:00+05:30",
  "location_match": true,
  "actual_atm_id": "ATM-HYD-00421",
  "notes": "Example demo outcome.",
  "feedback_label": "TRUE_POSITIVE"
}
```

Keep outcome labels explicit because the deck proposes feedback/evaluation and continuous improvement. The frontend does not perform evaluation; it records the operational outcome that can later be consumed by evaluation services.

16. Audit Event Contract

```
{
  "event_id": "AUD-501991",
  "timestamp": "2026-09-16T13:44:02+05:30",
  "actor_role": "LEA",
  "actor_display": "Officer 102",
  "event_type": "ALERT_ACKNOWLEDGED",
  "entity_type": "ALERT",
  "entity_id": "AL-20481",
  "summary": "Alert acknowledged by assigned LEA unit.",
  "ip_or_device_reference": "OPTIONAL-MASKED"
}
```

17. I4C Command Center — Screen Specification

Region	Data / component	Frontend behaviour
Top KPIs	Active cases; high-risk predictions; active alerts; alerts awaiting acknowledgement; actions in progress.	Clickable KPI cards filter the main workspace.
Main GIS	District risk polygons + ATM overview dots.	District click opens drill-down; filters update map and list together.
Alert queue	Alert ID, district, risk, predicted window, status, responsible LEA/bank.	Click opens alert detail.
Coordination	LEA acknowledgement and bank response state.	Show gaps/overdue states; allow authorized coordination action.
Case search	Case ID / complaint ID / ATM ID / district.	Global search opens authorized detail.
Activity	Recent audit events.	Click opens audit timeline.

I4C homepage wireframe concept

Header: NIRIKSHAK • Command Center • current scope • notifications • user

Row 1: Active Cases | High-Risk Predictions | Active Alerts | Pending Actions

Main body: large Telangana map on left/center + live alert queue on right.

Bottom: district summary / coordination status / recent activity.

18. LEA Investigation Console — Screen Specification

Screen	Primary content
Dashboard	Cases, high-risk predictions, active alerts, assigned actions, mini GIS.
Cases	Case list with status, crime category, reported amount, latest intelligence time, risk.
Case Detail	Complaint summary, prediction card, map, path, alert history, action timeline.
Prediction Detail	Predicted ATM, window, score/level, supporting paths, explanation factors, freshness.
GIS	Telangana/district/area/ATM drill-down with case-linked risk.
Investigation	Notes, assignments, evidence metadata, alert actions and outcome.

19. Bank/FI Console — Screen Specification

Screen	Primary content	Actions
Dashboard	Incoming alerts; pending acknowledgements; active responses; affected cases.	Open queue / filter.
Response Queue	Alert ID; account/transaction reference where authorized; location; window; risk; status.	Acknowledge / open.
Alert Detail	Prediction summary; related case; location; relevant financial context; recipient status.	Acknowledge; verify; record response; escalate.
Response History	Previous actions, timestamps, actor and outcome.	View only / authorized edit.
Audit	Bank-side activity timeline.	Review.

20. Citizen / Victim View — Keep It Small

The citizen is an input/status stakeholder, not an intelligence consumer. The prototype should deliberately avoid exposing sensitive predicted locations, account chains or investigation details.

Screen	Fields / information
Complaint Form	Incident type; incident time; amount lost; transaction/reference details; victim contact; optional known location; acknowledgement/consent.
Complaint Confirmation	Complaint ID; submission time; acknowledgement; safe next steps.
Status	Received -> Under Investigation -> Financial Intelligence Processing -> Action Initiated -> Resolved/Closed.
Follow-up	Requests for additional information, if any.

21. Shared Alert Detail — Common Component

This component should be reusable in LEA, Bank and I4C views with role-specific fields/actions.

Section	Data
Header	Alert ID; severity; status; created time; expiry/window.
Why this alert?	Risk score/level; supporting path count; convergence count; freshness; explanation factors.
Where?	District; area; ATM; coordinates; map link.
When?	Predicted withdrawal start/end; last observed event.
Related case	Case ID; complaint ID; crime category.
Recipients	LEA / Bank / I4C delivery and acknowledgement states, according to role.
Actions	Acknowledge; assign; verify; notify; record action/outcome — only where authorized.

Section	Data
Audit	Chronological event timeline.

22. Frontend Data Flow

The frontend should consume domain objects rather than model-specific outputs.

- **1. Case creation:** complaint is stored and receives a case/complaint ID.
- **2. Intelligence:** prediction service publishes one or more active predictions linked to the case.
- **3. Spatial aggregation:** district/area/ATM risk objects are made available to GIS.
- **4. Alerting:** qualifying predictions create alert objects with recipients and expiry.
- **5. Role filtering:** each frontend view requests/receives only the authorized representation.
- **6. Human action:** user records acknowledgement, assignment, verification or other authorized action.
- **7. Outcome:** operational result is recorded and linked back to alert/case/prediction.
- **8. Audit:** every significant state/action change appears in the audit timeline.

23. Recommended API Resource Structure

Resource	Example frontend purpose
GET /cases	Case list / dashboard.
GET /cases/{caseId}	Case detail.
GET /cases/{caseId}/predictions	Prediction cards and history.
GET /predictions/{predictionId}	Prediction/intelligence detail.
GET /gis/telangana	District map + overview ATM points.
GET /gis/districts/{districtId}	Area polygons + ATM points.
GET /atms/{atmId}	ATM/location detail.
GET /atms/{atmId}/withdrawals	Previous withdrawal history.
GET /predictions/{predictionId}/paths	Predicted fund-flow paths.
GET /alerts	Role-filtered alert queue.
GET /alerts/{alertId}	Alert detail.
POST /alerts/{alertId}/acknowledge	Record acknowledgement.
POST /alerts/{alertId}/actions	Record operational action.
POST /alerts/{alertId}/outcome	Record outcome/feedback.
GET /audit	Audit timeline.
GET /notifications	Notification center.

These are suggested resource names for the prototype contract, not a claim about the final production API. The important requirement is stable domain-level payloads and identifiers.

24. Dummy Data Pack the Frontend Team Should Build Against

Until the integrated prediction/risk services are connected, create local JSON fixtures with these entities. Do not invent a different schema for the frontend.

Fixture	Minimum quantity for demo	Purpose
cases.json	8–12 cases	Dashboard, case list, case detail.
predictions.json	8–15 predictions	Risk cards, prediction detail, map links.
districts.geojson	Telangana districts	Actual district polygons.
areas.geojson	Selected Hyderabad/district areas	Area-level heat/risk polygons.
atms.json	30–100 ATM points	State dots and district drill-down.
withdrawals.json	50–150 records	ATM history screen.
paths.json	10–20 predicted paths	Fund-flow visualisation.
alerts.json	10–20 alerts	Alert queue and lifecycle.
actions.json	10–20 actions	LEA/bank response workflow.
audit.json	30+ events	Audit timeline.
notifications.json	15–25 items	Notification center.

Minimum golden-path demo data

At least one case must have a complete chain:

25. Recommended 3–5 Minute Judge Walkthrough

Time	Action	What the judge should understand
0:00–0:30	Open Nirikshak and show role-based login / LEA view.	This is a secure operational platform, not a generic dashboard.
0:30–1:00	Open a complaint/case.	The system starts from the cybercrime complaint.
1:00–1:45	Open prediction detail.	The system forecasts a likely cash-out location and time window.
1:45–2:30	Open Telangana map -> Hyderabad -> area.	Risk is spatially drillable, not just a single score.
2:30–3:00	Click ATM.	ATM has location, bank, coordinates, map link, history and related intelligence.
3:00–3:30	Open fund-flow path.	Prediction is linked back to the case and transaction path.
3:30–4:15	Open alert -> acknowledge -> show bank/I4C recipient status.	Prediction becomes coordinated actionable intelligence.
4:15–4:45	Record action and outcome.	The loop closes with human action and feedback.
4:45–5:00	Switch to I4C view.	I4C can coordinate across LEA/bank responses.

26. Frontend Build Order

Phase	Build	Exit condition
1	Common shell + RBAC + navigation	Each role reaches the correct dashboard.
2	Case + prediction detail	Golden case is coherent.
3	Telangana GIS overview	Real district polygons render; ATM dots appear.
4	District drill-down	Area polygons + ATM points render and filter.

Phase	Build	Exit condition
5	ATM detail	ATM risk/history/case links work.
6	Path visualisation	Observed/predicted edges are distinguishable.
7	Alert workflow	Alert lifecycle and acknowledgements work.
8	LEA actions/outcomes	Action and outcome are recorded.
9	Bank/I4C views	Role-specific coordination is visible.
10	Polish + walkthrough	One end-to-end golden path works without dead ends.

27. Non-Negotiable UI Rules

Rule	Reason
Observed and predicted information must be visually distinct.	Prevents the system from implying that a forecast is a confirmed event.
Every prediction must link to a case.	The PS is complaint/case-driven, not generic hotspot prediction.
Every predicted ATM must have a predicted time window when available.	The objective is proactive intervention, not only spatial ranking.
Every ATM detail must expose coordinates and an external map link.	Makes the prediction operationally locatable.
Area risk must use actual geographic polygons where possible.	Avoids a superficial radial 'AI heatmap' appearance.
State-level ATM dots are overview markers; district drill-down activates details.	Preserves map readability and matches the requested interaction.
District highlighting must be driven by returned aggregation values.	Keeps the UI independent of business logic.
Risk score and risk level must be shown together.	Provides both machine-readable and human-readable risk.
Alerts need lifecycle status and timestamps.	Shows real operational workflow.
Actions and outcomes must be recorded.	Closes the prediction -> intervention loop.
Do not expose sensitive intelligence to citizens or unauthorized roles.	Supports realistic RBAC and security posture.
Avoid 20+ charts.	The judges need the operational story, not analytics clutter.

28. What Should NOT Be Sent to Frontend

- Raw model internals or model-specific tensors/embeddings.
- Unnecessary sensitive account identifiers.
- Full transaction data when a role only needs the relevant subset.
- Unexplained internal scoring components that cannot be interpreted by investigators.
- Duplicate versions of the same prediction without status/version metadata.
- Synthetic-data labels that could accidentally make the UI look like real government operational data.

If dummy data is used in the prototype, label the environment/data appropriately in a subtle but visible way such as DEMO / SIMULATED DATA. Never present synthetic records as actual NCRP/I4C records.

29. Frontend Integration Checklist

Check	Done when...
Role-based navigation	Citizen, LEA, Bank/FI and I4C see different authorized views.
Case linkage	Every prediction/alert shown in the demo links back to a case.
GIS state	Telangana districts render from geographic data.
District drill-down	Clicking a district reveals area risk polygons and ATM details.
ATM interaction	ATM opens a detail panel with risk, bank, coordinates and history.
Map link	Coordinates produce a clickable external Google Maps location.
Prediction detail	Risk, window, freshness and supporting intelligence are visible.
Path	Victim -> account -> account -> predicted ATM is visually understandable.
Alert lifecycle	Generated -> delivered -> acknowledged -> action -> outcome can be demonstrated.
Bank response	Bank can acknowledge and record response in the demo.
I4C coordination	I4C can see recipient/action status.
Audit	Key actions appear chronologically.
Observed vs predicted	Visual language clearly separates them.
Data consistency	The same case/ATM/alert IDs remain consistent across all screens.
No dead-end screens	Every major card/button in the walkthrough has a meaningful destination.

30. Final Product Definition

Nirikshak is not the map. The map is the spatial interface to the intelligence.

Nirikshak is not the alert. The alert is the mechanism that converts a prediction into coordinated attention.

Nirikshak is not the ML output. The prediction is useful only when it is tied to a complaint, location, time window, supporting path, authorized recipient, human action and recorded outcome.

The uploaded deck already contains the predictive, geospatial, alerting and stakeholder foundation. The frontend should now make that foundation tangible through one coherent operational workflow.

END OF SPECIFICATION